

HDOC DAY-24

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Batch-15

Question:

Write a C program that displays a menu and uses a switch statement to print the “Increasing patterns” selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameterless function with no return value.

Aim

To write a C program using a switch statement and nested loops to display different increasing patterns selected by the user.

Algorithm

1. Start.
2. Display the menu containing the **5 increasing patterns**.
3. Input the user's **choice**.
4. Use a switch statement to check the choice.
5. According to the choice, call the corresponding pattern function.
6. Input the number of rows n inside the selected function.
7. Use **nested loops** to print the selected pattern.
8. If the choice is not from 1 to 5, display "**Invalid choice**".
9. Stop.

>C program:

```
File Edit Search View Document Help
1 #include <stdio.h>
2
3 void pattern1();
4 void pattern2();
5 void pattern3();
6 void pattern4();
7 void pattern5();
8
9 int main()
10 {
11     int choice;
12
13     printf("1. Increasing right-angled triangle\n");
14     printf("2. Increasing row-number triangle\n");
15     printf("3. Increasing natural-number pattern\n");
16     printf("4. Consecutive-number triangle\n");
17     printf("5. Increasing alphabet pattern\n");
18
19     printf("Enter your choice: ");
20     scanf("%d", &choice);
21
22     switch(choice)
23     {
24         case 1: pattern1(); break;
25         case 2: pattern2(); break;
26         case 3: pattern3(); break;
27         case 4: pattern4(); break;
28         case 5: pattern5(); break;
29         default: printf("Invalid choice");
30     }
31
32     return 0;
33 }
34
35 void pattern1()
36 {
37     int n,i,j;
38     printf("Enter number of rows: ");
39     scanf("%d", &n);
40     for(i=1; i<=n; i++)
41     {
42         for(j=1; j<=i; j++)
43             printf("%d ", j);
44         printf("\n");
45     }
46 }
47
48 void pattern2()
49 {
50     int n,i,j;
51     printf("Enter number of rows: ");
52     scanf("%d", &n);
53 }
```

```
File Edit Search View Document Help
48 void pattern2()
49 {
50     int n,i,j;
51     printf("Enter number of rows: ");
52     scanf("%d", &n);
53     for(i=1; i<=n; i++)
54     {
55         for(j=1; j<=i; j++)
56             printf("%d ", i);
57         printf("\n");
58     }
59 }
60
61 void pattern3()
62 {
63     int n,i,j;
64     printf("Enter number of rows: ");
65     scanf("%d", &n);
66     for(i=1; i<=n; i++)
67     {
68         for(j=1; j<=i; j++)
69             printf("%d ", j);
70         printf("\n");
71     }
72 }
73
74 void pattern4()
75 {
76     int n,i,j,num=1;
77     printf("Enter number of rows: ");
78     scanf("%d", &n);
79     for(i=1; i<=n; i++)
80     {
81         for(j=1; j<=i; j++)
82             printf("%d ", num++);
83         printf("\n");
84     }
85 }
86
87 void pattern5()
88 {
89     int n,i,j;
90     printf("Enter number of rows: ");
91     scanf("%d", &n);
92     for(i=1; i<=n; i++)
93     {
94         for(j=1; j<=i; j++)
95             printf("%c ", 'A'+j-1);
96         printf("\n");
97     }
98 }
99
```

Screenshot taken
View image

>Output

```
Session Actions Edit View Help
priyadarshna@kali: ~
$ gcc day24.c -o day24
priyadarshna@kali: ~
$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 1
Enter number of rows: 4
*
*
*
*
priyadarshna@kali: ~
$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter number of rows: 4
1
2 2
3 3 3
4 4 4 4
priyadarshna@kali: ~
$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter number of rows: 4
1
1 2
1 2 3
1 2 3 4
priyadarshna@kali: ~
$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 4
Enter number of rows: 4
1
2 3
4 5 6
7 8 9 10
priyadarshna@kali: ~
$
```

>

```
Session Actions Edit View Help
priyadarshna@kali: ~
$ ./day24
*
*
*
*
priyadarshna@kali: ~
$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter number of rows: 4
1
2 2
3 3 3
4 4 4 4
priyadarshna@kali: ~
$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter number of rows: 4
1
1 2
1 2 3
1 2 3 4
priyadarshna@kali: ~
$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 4
Enter number of rows: 4
1
2 3
4 5 6
7 8 9 10
priyadarshna@kali: ~
$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 5
Enter number of rows: 4
A
A B
A B C
A B C D
priyadarshna@kali: ~
$
```